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Perspective

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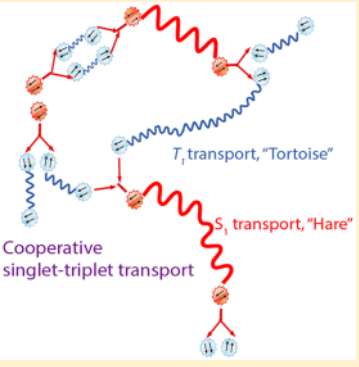
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Exciton Transport in Singlet Fission Materials: A New Hare and Tortoise Story

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ABSTRACT: Singlet fission is promising for redistributing the solar spectrum to overcome the Shockley−Queisser limit for single-junction solar cells using molecular materials. Despite recent experimental and theoretical advances in understanding the underlying mechanisms, how exciton transport is coupled to singlet fission dynamics is much less explored. In this Perspective, we examine exciton transport in singlet fission materials, highlighting the use of transient absorption microscopy (TAM) to track the population of different states in both spatial and temporal domains. In contrast to the conventional picture where singlet and triplet excitons migrate independently, TAM measurements of acene single crystals reveal cooperative transport between fast-moving singlet and slow-moving triplet excitons. Such cooperative transport is unique to singlet fission materials and allows hundreds of nanometers triplet migration on the nanosecond time scale, beneficial for solar cell applications. The transport of triplet pair intermediates and general criteria for achieving cooperative singlet−triplet transport are also discussed.



Photovoltaics convert clean, abundant, and sustainable solar energy into electrical power. The energy conversion efficiency for a single-junction solar cell is thermodynamically limited by the Shockley−Quiesser limit;[1] for instance, this limit is 33.7% for silicon solar cells. Solar energy below the bandgap of light-absorbing material is not absorbed, and incident photons with energy higher than the bandgap are dissipated as heat. Photon downconversion in excitonic materials are promising to redistribute the solar spectrum to overcome the Shockley−Quiesser limit. As the name suggests, down-conversion occurs when an exciton created by a high-energy photon is converted to two or more lower-energy excitons. Singlet fission is a downconversion process in certain organic semiconductors that splits a singlet (spin 0) exciton into two triplet (spin 1) excitons, each with roughly half of the singlet energy, generating one extra electron−hole pair per photon absorbed (Figure 1), which would have been otherwise wasted via thermalization.[2,3] Singlet fission is a spin-allowed process, and it is now generally accepted that the conversion from the singlet to triplet excitons involves two intermediates, ^{1}[TT] and ^{1}[T···T],[4−6] which are doubly excited and spin-correlated triplet pair states with overall singlet character (shown in Figure 1). ^{1}[TT] can be separated physically by triplet−triplet energy transfer to form a spatially separated yet spin-entangled ^{1}[T···T] state.[4,7] ^{1}[T···T] then eventually decoheres into two free triplet excitons T_1 by spin dephasing[4,7](Figure 1). Excitations in certain molecular crystals, polycrystalline and disordered thin films, dimers, polymers, as well as molecular aggregates have been demonstrated to undergo efficient singlet fission to form triplet excitons.[8−20] Efficiency of solar cells based on singlet fission materials coupled to silicon in principle could be increased to 45%.[3] Triplet yield of 200% and external quantum efficiency above 100% have been achieved in singlet fission-based photovoltaics.[21,22]

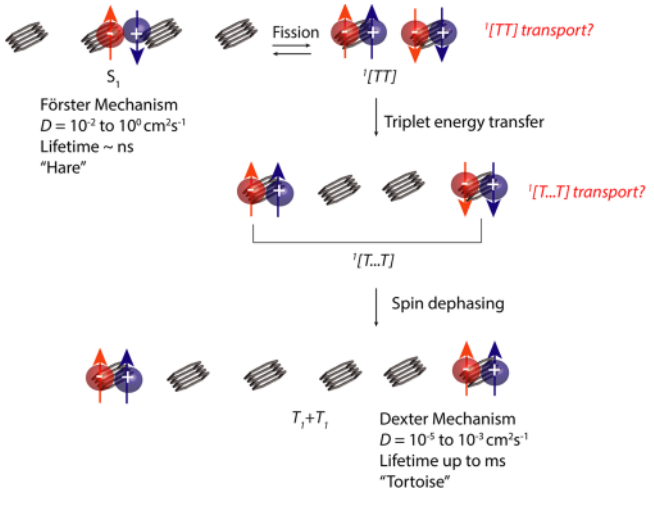


Figure 1. Schematic illustration of how the singlet fission process coupled with singlet, triplet, and intermediate states transport.

Understanding and controlling exciton transport is a crucial aspect for realizing singlet fission as a viable technology. In order for the conversion from light to electricity to occur, the energy carriers have to migrate efficiently to the donor−acceptor interface for charge separation. Specifically, for singlet fission-based bilayer photovoltaic cells, the main goal is to achieve long-range triplet exciton transport as opposed to singlet excitons for charge buildup at the interface. The low diffusion constant of triplet excitons is an obstacle in achieving

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long-range transport, and the thickness limited by triplet diffusion length might not be sufficient for light absorption. For instance, the optimal active layer thickness of the pentacene-based bilayer device is limited to 10−15 nm, far from the thickness required for sufficient absorption for practical single-junction solar cells.[23]

A major obstacle in measuring singlet exciton and triplet exciton migration lies in the complex excited-state population dynamics involved in singlet fission.

Despite the significant research efforts that have been devoted to investigating singlet fission dynamics and designing structures for high triplet yield, much less emphasis has been placed on exciton transport in these materials. Readers interested in the mechanistic details of singlet fission dynamics and device studies are directed to these review articles.[3,24−26] A major obstacle in measuring singlet exciton and triplet exciton migration lies in the complex excited-state population dynamics involved in singlet fission. In addition, questions remain open for transport of the intermediate triplet pairs that are potentially useful for multielectron transfer reactions.[7,27−35] Thus, experimental tools that are capable of tracking both spatially and temporally the population of the singlet, triplet, and intermediate states would be ideal.

In this Perspective, we discuss recent progress in the understanding of exciton transport in singlet fission materials. A key point to realize is that singlet and triplet exciton transport in these materials cannot be treated independently because their populations are coupled through singlet fission dynamics. Recently, optical microscopy approaches have been employed to directly image exciton transport.[36−38] In particular, transient absorption microscopy (TAM) using excited-state absorption as the imaging contrast is capable of tracking both the bright singlet and the dark triplet population.[38,39] Importantly, the simultaneously high spatial and temporal resolutions allow for direct examination of the underlying mechanisms of exciton transport. We highlight in this Perspective that the migration of the slow-moving triplet excitons can be enhanced by that of the mobile singlet excitons, leading to long-range triplet transport in endothermic singlet fission materials. Experimental strategies to directly address the transport of the intermediate triplet pair states are also discussed. The design guidelines for singlet fission materials need to take coupled singlet−triplet transport into account in addition to optimizing singlet fission rates and triplet yields. We suggest delayed fluorescence to be employed as an indicator for cooperative singlet−triplet transport as regenerated singlet population also leads to delayed fluorescence. The singlet-mediated triplet transport pathways are particularly important for solar cell applications that employ polycrystalline thin films in which triplet lifetimes are limited by traps and defects to the ns time scale.

Brief Review of Singlet and Triplet Exciton Transport Mechanisms. Singlet fission occurring for acene molecular crystals, such as tetracene, have long been employed as model systems for Frenkel excitons.[40−42] Theory on singlet exciton transport in molecular solids was first developed by Holstein[43] and has been investigated since by many others.[40,44,45] Singlet and triplet excitons transport through different mechanisms due to the difference in the electronic coupling for the process. Singlet excitons migrate by Förster energy transfer mechanisms using Coulomb coupling from dipole−dipole interaction. Singlet diffusion constants usually range from 10^{-2} to 1 cm^2 s^{-1}. Larger transition dipole moments result in larger coupling and higher diffusion constants. On the other hand, the singlet exciton lifetime is generally around a few nanoseconds and decreases as the transition dipole moment increases. Because the exciton diffusion length is given by $\sqrt{D\tau}$, where D is the diffusion constant and τ is the lifetime, there exist fundamental upper bounds for singlet exciton diffusion lengths, as detailed by Yost et al. in ref 45.

In contrast, triplet excitons are not able to migrate by Förster mechanism because the triplet excited state to singlet ground state transition is a spin-forbidden process with 0 transition dipole. Consequently, triplets can only transfer their energy through exchange coupling known as Dexter energy transfer, occurring through the exchange of two electrons between the donor and acceptor. Exchange coupling decays exponentially as a function of distance, and it is a much shorter-range interaction than the Coulomb coupling that scales with $\frac{1}{R^6}$. Yost et al.[45] computed singlet and triplet transport in tetracene, rubrene, and pentacene crystals, concluding triplet diffusion constants to be about 3 orders of magnitude lower than those of the singlet. However, the triplet lifetime is theoretically only limited by spin−orbit coupling and can be many orders of magnitude longer than that of the bright singlet state in single crystals with low defect density. Therefore, the long triplet lifetimes could potentially lead to longer diffusion lengths than the singlet excitons despite slower diffusion constants. As an analogy, singlet excitons migrate like the "hare" with fast speeds and short lifetimes, while triplet excitons behave as the "tortoise" with low diffusion constants and long lifetimes.

Experimentally, the singlet exciton diffusion length is most commonly measured indirectly by photoluminescence (PL) quenching.[46] However, these measurements do not test the fundamental assumptions of exciton transport, and therefore, they do not provide significant insights into mechanisms. Direct measurements of singlet exciton transport in single anthracene crystals was first demonstrated by Fayer and co-workers using picosecond transient grating experiments.[47] Singlet exciton diffusion was observed to accelerate as the temperature decreased, indicating that coherent effects due to exciton delocalization play a role. A diffusion constant of ∼2 cm^2s^{-1} was measured at 10 K for singlet excitons in anthracene. Triplet exciton transport was not discussed by Fayer and co-workers in ref 47 likely because singlet fission in anthracene is slow and with low triplet yield.

The main challenge in measuring triplet exciton transport is that the triplet state is optically dark and the majority of triplet diffusion measurements to date are indirect. For instance, Poletayev et al. estimated the triplet diffusion length in pentacene single crystals and evaporated thin films using a bilayer device structure. Triplet exciton diffusion under 1 sun illumination was estimated to be 40−80 nm in evaporated films and 350−800 nm in single crystals.[48] Najafov et al.[49] showed a triplet exciton diffusion length of up to 8 μm in rubrene single crystals by employing photoconductivity measurements. Exciton diffusion in rubrene single crystals was visualized by Irkhin et al. using localized photoexcitation and spatially resolved detection of delayed singlet fluores-

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cence.[36] A triplet exciton diffusion length of 4 μm was determined from the spatial exponential decay of the fluorescence that originated from singlet excitons regenerated by fusion and triplet−triplet annihilation.

> Experimental tools that are capable of tracking both spatially and temporally the population of the singlet, triplet, and intermediate states would be ideal.

Akselrod et al. studied triplet transport in tetracene single crystals and polycrystalline thin films by time-resolved PL microscopy and measured a triplet diffusion length up to 6 μm.[37] Measurements of the triplet exciton diffusion constant on the μs time scale were achieved by quantifying the broadening of profiles of the delayed fluorescence as a function of time (Figure 2a). The excitation beam at position (x_0, y_0) generates an initial population $n(x,y,0)$, which can be approximated as a two-dimensional (2D) Gaussian function.

$$n(x, y, 0) = N \exp\left[-\frac{(x - x_0)^2}{2{\sigma_{x,0}}^2} - \frac{(y - y_0)^2}{2{\sigma_{y,0}}^2}\right] \quad (1)$$

If the excitonic motion is diffusive, then the population as a function of space and time will be governed by a diffusion equation that includes decay to the ground state

$$\frac{\partial n(x, y, t)}{\partial t} = D\left[\frac{\partial n^2(x, y, t)}{\partial x^2} + \frac{\partial n^2(x, y, t)}{\partial y^2}\right] - \frac{n(x, y, t)}{\tau} \quad (2)$$

Here $n(x,y,t)$ is the exciton population at time t, D is the diffusion constant, and τ is the exciton lifetime. The solution to eq 2 shows that the population profile at any later time is also Gaussian, $n(x,y,t) = N \exp\left(-\frac{(x-x_0)^2}{2{\sigma_{x,t}}^2} - \frac{(y-y_0)^2}{2{\sigma_{y,t}}^2}\right)$, and the average distance that excitons travel in time t is defined as L and is given by

$${L_{x(y)}}^2 = {\sigma_{x(y),t}}^2 - {\sigma_{x(y),0}}^2 \quad (3)$$

For diffusive transport, L^2 grows linearly as a function of delay time, as given by ${L_{x(y)}}^2 = 2D_{x(y)}t$, and D is simply half of the slope. Diffusive triplet exciton transport with a diffusion constant of 1.35×10^{-3} cm^{2}s^{-1} on the μs time scale in single tetracene crystals was extracted from the time-dependent PL microscopy measurements (Figure 2b). Long-range μm triplet exciton diffusion in single-crystalline tetracene and rubrene from these PL-based measurements[36,37] has been attributed to the long triplet lifetime that compensates the moderate diffusion constant resulting from the Dexter mechanism. The relationship between singlet fission dynamics and exciton transport was not explored in detail.

Coupled Singlet−Triplet Exciton Transport in Acene Single Crystals. For singlet fission materials, singlet and triplet population are not independent but related by singlet fission dynamics. Indeed, one of the hallmarks of singlet fission in tetracene and rubrene is the delayed singlet fluorescence from triplet recombination. Therefore, singlet and triplet exciton transport should not be treated independently, but rather how their population coupled via singlet fission has to be considered. In our recent works, we have employed TAM to track exciton population in both spatial and temporal domains to elucidate the interplay between exciton transport and singlet fission.[38,39] TAM has two main advantages over the time-resolved PL microscopy method. First, excited-state absorption is employed as imaging contrast in TAM, which allows for the imaging of singlet and triplet exciton population simultaneously. Second, the temporal resolution of ~300 fs for TAM is much better than that of ~100 ps offered by PL, sufficient for detailed investigation of fast singlet fission dynamics. One potential drawback of TAM is the higher pump intensity required compared to PL, and therefore, exciton−exciton annihilation needed to be taken into account when modeling the results.

To image exciton transport using TAM, the pump beam is held at a fixed position while the probe beam is scanned relative to the pump with a Galvanometer scanner, and the pump-induced change in probe transmission (ΔT) is plotted as a function of probe position to form an image (Figure 3).[38] The probe wavelengths to monitor singlet and triplet excitons are chosen based on their excited-state spectra. As an example, Figure 3a shows that the decay-associated spectrum (DAS) of the excited singlet state with a lifetime of <100 ps is mostly in the visible spectral region, distinct from that of the long-lived

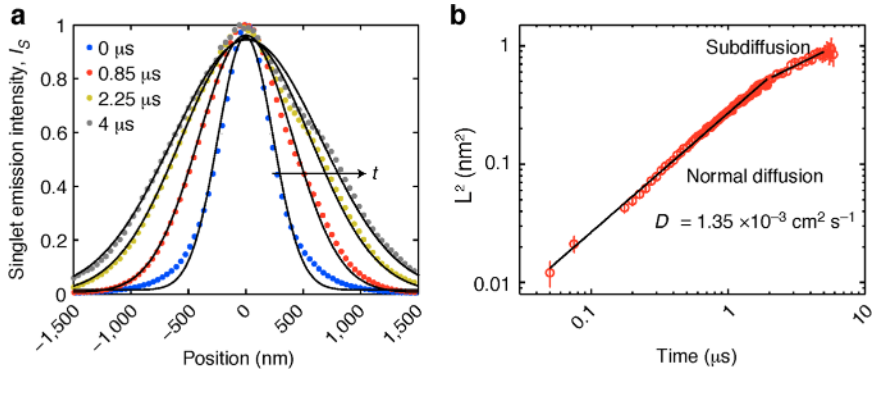


Figure 2. Triplet exciton transport in tetracene by time-resolved PL microscopy. (a) Cross sections of the emission intensity map at four time points showing spatial broadening of the intensity distribution of the delay singlet florescence fitted to Gaussian functions. (b) Time evolution of the mean-square displacement of triplet excitons showing transition from diffusive to subdiffusive transport. Adapted with permission from ref 37. Copyright 2014 Nature Publishing Group.

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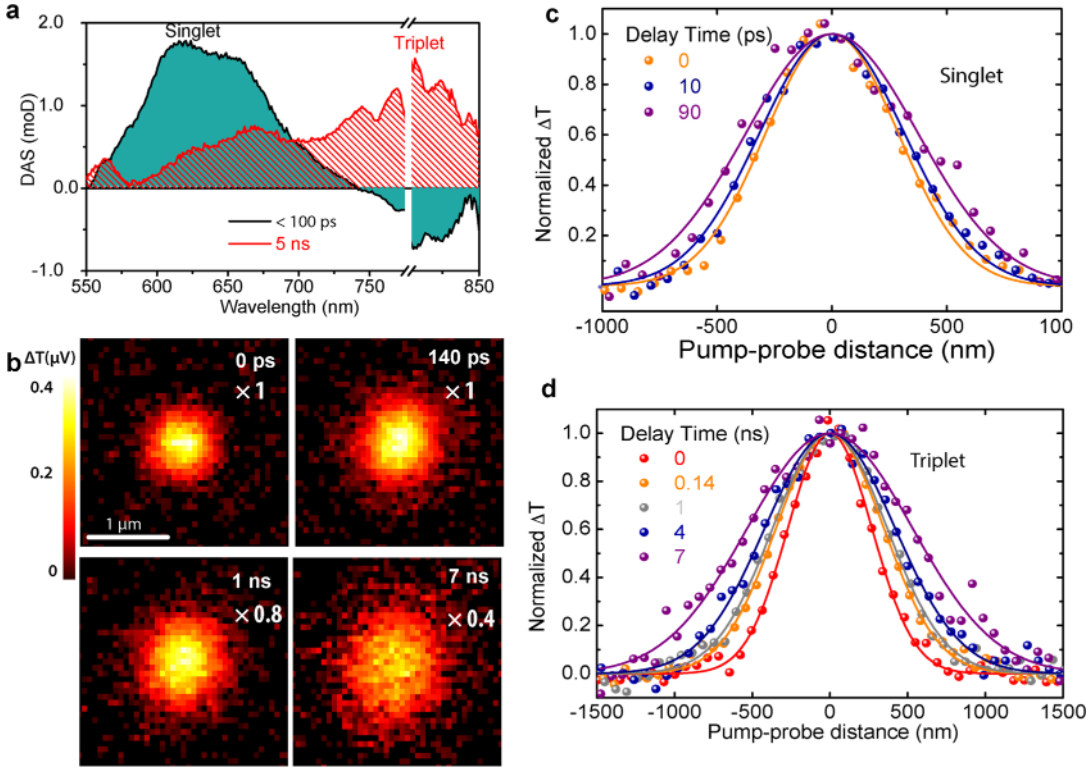


Figure 3. Direct imaging of tetracene singlet and triplet exciton transport by TAM. (a) DAS of the triplet and singlet excitons obtained by global analysis of the ensemble transient absorption spectra. (b) Triplet exciton propagation in the $a-b$ plane imaged with a probe wavelength of 810 nm. The images show the spatial distribution of the ΔT signal measured at the indicated pump−probe delay times. (c,d) Cross sections of the singlet (probed at 630 nm) and triplet (probed at 810 nm) transport TAM images along the b axis fitted with Gaussian functions with the maximum ΔT signal normalized. Adapted from ref 38. Copyright 2015 Macmillan Publishers Ltd.

triplet state in the near-infrared (NIR) for tetracene single crystals.[38] The lifetime of the singlet excitons is limited by a singlet fission process. Note that the negative feature in the NIR region of the singlet spectra is due to the conversion from singlet to triplet population rather than stimulated emission. Examples of TAM images of singlet (probed at 630 nm) and triplet exciton (probed at 810 nm) transport in a single tetracene crystal are shown in Figure 3. The exciton transport processes are investigated in the $a-b$ plane, and ΔT is integrated over the c axis.[38] Transport along the b axis is more rapid than that along the a axis due to stronger intermolecular coupling, leading to elliptical exciton population images at long delay times. We fit the time-dependent exciton population profiles along the b axis to Gaussian functions (Figure 3c,d) using similar method as those described in ref 37. A singlet exciton diffusion constant D_S of 2.8 cm^2 s^{-1} is extracted from the TAM results for tetracene.

The design guidelines for singlet fission materials need to take coupled singlet−triplet transport into account in addition to optimizing singlet fission rates and triplet yields.

Most interestingly, triplet transport in tetracene on the ns time scale deviates significantly from diffusive transport. L^2 is highly nonlinear as a function of delay time, and triplet excitons migrate at a much faster speed on the subns to ns time scales (Figure 4a) than the diffusive triplet transport on the μs time scale shown in Figure 2b. Through kinetic modeling, we have shown that the nondiffusive and enhanced triplet transport on the ns time scale resulted from singlet−triplet exciton population interconversion. The exciton population interconversion can be described by the following four excited-state processes

$$S_1 \xrightarrow{k_{\text{fission}}} {}^1[TT] \xrightarrow{\text{triplet transfer}} {}^1[T\cdots T] \quad (4)$$

$${}^1[T\cdots T] \xrightarrow{k_{\text{fusion}}} S_1 \quad (5)$$

$${}^1[T\cdots T] \xrightarrow{k_{\text{dissociation}}} 2T_1 \quad (6)$$

$$2T_1 \xrightarrow{k_{\text{annh_TT}}} S_1 + S_0 \quad (7)$$

The coupled diffusion equations that take into account the population conversion successfully explain the TAM results (Figure 4a).[38,39] The cooperative transport is summarized as follows, and more details of the modeling can be found in our recent publications.[38,39] Because of the ability of triplet excitons to regenerate singlet excitons, these excitations can migrate as singlet excitons on time scales long after the initial step of singlet fission completes. Despite the fact that these excitations only spend a brief time in the singlet state before fission occurs again, the small singlet population can enhance triplet transport significantly because the singlet excitons move at a diffusion constant 3 orders of magnitude larger than that of the triplet. Figure 4b compares the singlet-mediated con-

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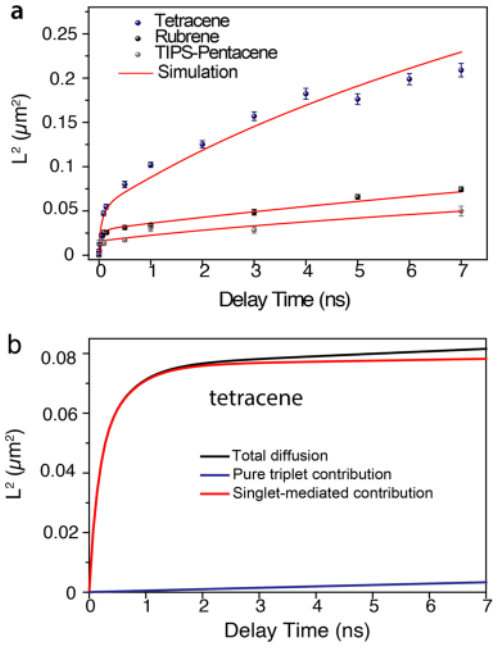


Figure 4. Triplet transport in tetracene, rubrene, and TIPS-pentacene. (a) Experimental and simulated time evolution of the spatial profile $L^2 = \sigma(t)^2 - \sigma(0)^2$ for tetracene, rubrene, and TIPS-pentacene when probing triplet. (b) Simulated contribution of singlet-mediated and pure triplet diffusion to L^2 along the fast transport axis for the ns time scale under solar fluences for tetracene. Adapted with permission from ref 38. Copyright 2016 Wiley-VCH.

tribution vs the contribution from triplet-only diffusion using the triplet diffusion constant reported from ref 37 in tetracene under solar fluences. The singlet-mediated contribution dominates in the ns and sub-ns time scales, while for the μs time scale, the probability of regenerating singlet decreases, and the triplet transport approaches the intrinsic diffusion constant.

The cooperative transport pathways are summarized in Figure 5. Once singlet fission occurs, the triplet exciton pair can either fuse back to one singlet before diffusing away or dissociate into two free triplets. At the sub-ns time scale, the effective triplet exciton transport is slower than the singlet diffusion constant D_S but is significantly larger than the intrinsic triplet diffusion constant D_T. This regime corresponds to the turning point in Figure 4a of the population profile. After triplet pair dissociation, the free triplet excitons have diffused away from their origins (pathway 2 in Figure 5). When two triplet excitons of different origins encounter each other, triplet−triplet annihilation can also lead to generation of the singlet exciton, which again accelerates triplet diffusion. For this time scale, the population of the singlet is quadratic with pump intensity because triplet−triplet annihilation is a bimolecular process. This picture differs from the conventional picture where triplets themselves diffuse over relatively long distances due to their long lifetimes and moderate diffusion constants and provides crucial details about early time exciton transport to states that are not accessible by the previous PL-based studies.

A key feature of exciton transport in singlet fission materials is the population interconversion between singlet and triplet states.

The rates of singlet fission and singlet regeneration determine the contribution from singlet-mediated triplet transport. The singlet regeneration rate needs to be comparable to the singlet fission rate to maintain a small singlet population on the ns time scale. We have compared singlet and triplet exciton diffusion and dynamics in tetracene, rubrene, and 6,13-bis(triisopropylsilylethynyl) pentacene (TIPS-pentacene) single crystals, a series of materials that have all demonstrated efficient singlet fission but with different singlet and triplet energy levels (Figure 4a).[39] Singlet fission in tetracene is the most endothermic among the three crystals, with a driving force $DE_{SF} = E(S_1) - 2E(T_1)$ of $-(0.1-0.2)$ eV, followed by rubrene with a DE_{SF} of ~ -0.05 eV.[3,23] The singlet energy is close to degenerate with twice the triplet energy in TIPS-pentacene ($DE_{SF} \approx 0$).[50] Triplet exciton population migrates at very different rates for the three crystals despite the fact that the triplet exciton diffusion rates in the three semiconductors have been calculated to be of the same order.[39] For the ns time scale, triplet excitons migrate most rapidly in tetracene, followed by rubrene, and the slowest in TIPS-pentacene (Figure 4a). The difference in triplet transport can be understood by the singlet fission and singlet regeneration rates, as summarized in Table 1. Because singlet fission is most endothermic in tetracene, the reverse process of triplet fusion should be most exothermic, which agrees well with the fact that the fission rate is lowest while the fusion rate constant is greatest in tetracene. The triplet fusion rate

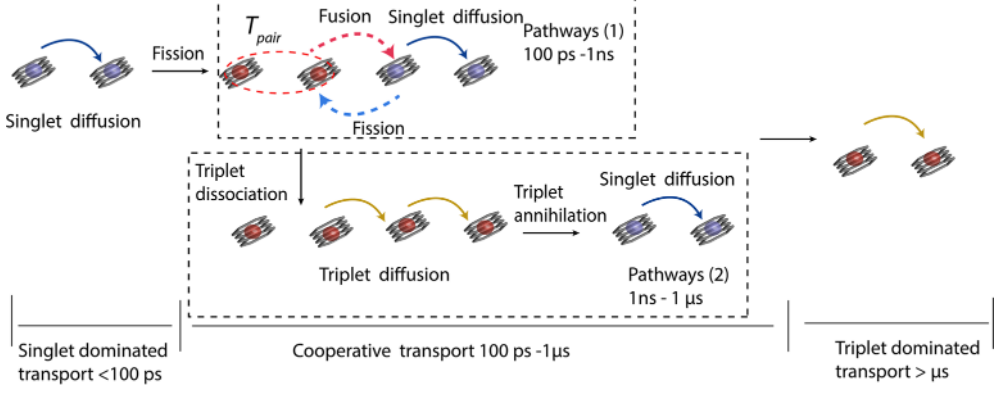


Figure 5. Schematics of different regimes of exciton transport in tetracene crystals. Adapted from ref 38. Copyright 2015 Macmillan Publishers Ltd.

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Table 1. Key Fitting Parameters Extracted for Tetracene, Rubrene, and TIPS-Pentacene

parameters	tetracene	rubrene	TIPS-pentacene
D_T (fast) intrinsic triplet diffusion constant along the fast axis	0.00228 $cm^2\ s^{-1}$	0.0035 $cm^2\ s^{-1}$	0.005 $cm^2\ s^{-1}$
D_S (fast) intrinsic singlet diffusion constant along the fast axis	2.8 $cm^2 s^{-1}$	2.3 $cm^2 s^{-1}$	2.8 $cm^2 s^{-1}$
fission rate	1/(120 ps)	1/(70 ps)	1/(5 ps)
fusion rate	1/(1 ns)	1/(1.2 ns)	1/(20 ns)
triplet−triplet annihilation	1.8×10^{-11} $cm^3\ s^{-1}$	5.1×10^{-12} $cm^3\ s^{-1}$	4.2×10^{-12} $cm^3\ s^{-1}$

constant is more than 10 times greater in tetracene and rubrene than that in TIPS-pentacene, leading to a much more active singlet regeneration (Table 1). This is consistent with that tetracene and rubrene exhibit more significant delayed fluorescence than TIPS-pentacene.

The TAM measurements suggest that systems with small endothermic gaps such as tetracene and rubrene are highly attractive when the triplet exciton diffusion length is added as a criterion in selecting singlet fission materials. The majority of the works in the field have been focusing solely on optimizing triplet yields and singlet fission rates. As a consequence, exothermic singlet fission, with energy gap $DE_{SF} = E(S_1) - 2E(T_1) > 0$ is usually preferred over the endothermic energetics. However, the exothermic singlet fission materials such as pentacene generally suffer from relatively short triplet diffusion lengths. A significant advantage for systems with small endothermic gaps is that long-range triplet exciton migration can be achieved in a ns time scale, much shorter than the previously expected μs time scale. For instance, the effective transport length L in 7 ns for triplets in tetracene is 310 nm under solar fluences (Figure 4b). Without the singlet-mediated contribution (i.e., without the ability to transport within the regenerated singlet state), the time required for a triplet exciton to diffuse the same distance would be 340 ns, almost 50 times longer. Singlet-mediated contribution is particularly useful for solar cell applications that employ polycrystalline thin films in which triplet lifetimes are limited by existence of static traps to be on the ns time scales. We have simulated the optimal device thickness for pentacene, TIPS-pentacene, rubrene, and tetracene assuming a triplet lifetime of 10 ns in our recent publication.[39] Due to the short exciton diffusion length, the optimal device thickness for pentacene is around 10−25 nm. In comparison, long-range triplet transport facilitated by singlet-mediated pathways allows for much thicker optimal device thickness of 150 and 260 nm for rubrene and tetracene, respectively. The singlet-mediated long-range triplet exciton in tetracene and rubrene is not entirely free. The increased triplet diffusion is matched by the increase in regenerated singlet exciton population and therefore leads to decreased triplet yield. Cooperative transport in tetracene costs a ∼20% decrease in triplet yield, which leaves a triplet yield of ∼180%.

The potential long-range triplet transport in addition to photostability and large absorption cross section makes PDI derivatives excellent candidates for singlet fission-based solar cells.

Transport of the Intermediate Triplet Pairs. As discussed above, a key feature of exciton transport in singlet fission materials is the population interconversion between singlet and triplet states. The intermediate triplet pairs $^1[TT]$ and $^1[T\cdots T]$ play an important role in the population interconversion. Note that $^1[TT]$ and $^1[T\cdots T]$ should be two different electronic states because the orbital overlap and interaction between the two triplet excitons are significantly reduced in $^1[T\cdots T]$,[4,6,32] even though many of the current works in the field do not make the distinction between them. Electronic coherence is lost but spin coherence is retained in $^1[T\cdots T]$. There is considerable recent interest in multielectron transfer as an alternative strategy to extract two excited electrons directly from the $^1[TT]$ state.[30,34] Understanding of the transport of $^1[TT]$ would be key for such application. Recent experiments on TIPS-pentacene nanoparticles suggest that the diffusion of $^1[TT]$ to disordered sites is responsible for their dissociation to isolated triplets.[35] The nature of these two intermediate states, especially $^1[TT]$, is currently under debate and has attracted significant research interests.[5,7,13,27−33,51−61] Recently, the understanding of $^1[TT]$ has been addressed by using advanced ultrafast spectroscopic techniques and by employing covalently linked dimers in solution where the $^1[TT]$ state does not dissociate.[16,19,30,31,33,62,63] Specifically, the electronic and vibronic coupling between $^1[TT]$ and S_1 states could be significant, which has been proposed to explain the rapid initial step in exciton fission as well as the emission directly from $^1[TT]$.[28,29,53,56,60,64]

$^1[TT]$ is likely not a strictly dark state, as suggested by its emission, resulting from the oscillator strength borrowing from the S_1 state.[29,56] Thus, $^1[TT]$ transport might not be limited by

Table 2. Singlet Fission Compounds with Small Edothermic Gaps

	E(S1) eV	E(T1) eV	ref
PDI	2.34−2.36	1.1−1.28	Ford *J. Phys. Chem.* **1987**, *91*, 6373 (experimental)
			Eaton; et al. *J. Am. Chem. Soc.* **2013**, *135*, 14701−14712 (experimental)
			Fukuzumi; et al. *J. Phys. Chem. A* **2008**, *112*, 10744−10752 (experimental)
			Le; et al. *J. Am. Chem. Soc.* **2018**, *140*, 814−826 (experimental)
o-xylylene derivative	2.7	1.4	Paci; et al. *J. Am. Chem. Soc.* **2006**, *128*, 16546−16553 (computational)
			Johnson; et al. *J. Am. Chem. Soc.* **2010**, *132*, 16302−16303 (experimental)
			Smith; et al. *Chem. Rev.* **2010**, *110*, 6891−6936
tetracyanoquin-odimethane (TCNQ) derivatives	1.94−2.18	1.08−1.22	Frankevich; et al. *Chem. Phys. Lett.* **1991**, *177*, 283 (experimental)
			Agostini; et al. *Chem. Phys.* **1993**, *173*, 177 (experimental)
			Smith; et al. *Chem. Rev.* **2010**, *110*, 6891−6936

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the slow Dexter mechanism. This could be beneficial for multielectron transfer applications. The difficulty in imaging transport of the triplet pairs arises from congested spectra with overlapping signature from all states including S_1, T_1, $^1[TT]$, and $^1[T\cdots T]$. The strong interaction between the two triplet excitons in $^1[TT]$ with a binding energy is as large as 100 meV,[27,30,33] and the dissociation of $^1[TT]$ to $^1[T\cdots T]$ should be thermally activated. Therefore, temperature-dependent measurements could be used to differentiate the transport of the two states. Even though we have not differentiated $^1[TT]$ and $^1[T\cdots T]$ in our initial studies,[38,39] it is possible to perform TAM at low temperatures. Another strategy is to employ vibrational probes such as IR absorption or Raman scattering instead of transient absorption as the imaging contrast. Asbury and co-workers have observed new electronic transitions in the mid-IR spectral range for $^1[TT]$ using ultrafast mid-IR transient absorption spectroscopy.[32] Frontiera and co-workers have demonstrated ultrafast Raman microscopy to identify vibrational modes unique to $^1[TT]$.[61] Exciton diffusion imaging could be achieved by spatially separating the pump and the mid-IR probe or the Raman probe similar to the TAM measurements.

Implications for Other Singlet Fission Materials. Singlet-mediated triplet transport pathways should also exist in other singlet fission materials with small endothermic gaps. Table 2 lists the systems likely to display cooperative transport based on reported singlet and triplet energy levels. Experimentally, delayed fluorescence can be used as an indicator for cooperative transport because it also resulted from regenerated singlet excitons. A particularly interesting candidate is the family of perylenediimide (PDI) molecules. PDIs are well-suited for photovaltaic applications because of their excellent photostability, large absorption cross section, and high charge mobility. Another attractive aspect is that molecular packing with different degrees of slip-stacking can be achieved by using different functional groups at the imide and bridge positions to modulate singlet fission rates. Singlet fission in mutliple PDI thin films with slip-stack geometries have been demonstrated.[14,20] As recently reported by Roberts and co-workers,[20] a maximal triplet production yield of 178% was achieved.

PDIs possess overall similar energetics as tetracene (Table 2), and certain derivatives also have comparable singlet fission and triplet fusion rates to tetracene (Table 3). Therefore, it is expected that singlet-mediated triplet transport should exist in PDI derivatives such as *N*,*N*′-bis(2-phenylethyl)-3,4,9,10-PDI (EP-PDI in ref 20). Indeed delayed fluorescence has been observed suggesting singlet−triplet population interconversion.[14,20] In addition, PDI thin films adopting a slip-stacked packing geometry have been shown to form more delocalized singlet excitons,[65] indicating mobile singlet excitons. A singlet diffusion length up to 2.5 μm was reported for a PDI thin film.[66] The cooperative exciton transport will be of particular importance in achieving long-range triplet exciton migration because triplet lifetimes are only on the order of tens of nanoseconds for PDIs[20] and the intrinsic triplet transport length would not be sufficient. The potential long-range triplet transport in addition to photostability and large absorption cross section makes PDI derivatives excellent candidates for singlet fission-based solar cells.

Table 3. PDI Derivatives That Likely to Display Cooperative Transport

molecule	E(S1) (eV)	E(T1) (eV)	DE_{SF} (meV)	fission rate	fusion rate
MO-PDI[20]	2.36	1.19	−270	1/(230 ps)	1/(1.04 ns)
EP-PDI[20]	2.36	1.19	−270	1/(260 ps)	1/(1.05 ns)
C3-PDI[20]	2.36	1.19	−250	1/(600 ps)	1/(1.44 ns)
tetraph-PDI[14]	2.08	1.14	−200	1/(180 ps)	1/(2 ns)

Besides singlet fission materials, cooperative transport could also exist in other systems where singlet and triplet exciton excitons interconvert. For certain inorganic semiconductors, such as halide perovskite quantum dots[67] and monolayer WS_2,[68] the energy gap between the singlet and the triplet exciton states is small, leading to population equilibrium at room temperature. In such systems, the triplet state can function as a long-lived reservior to support long-range exciton transport.

Cooperative single−triplet transport makes singlet fission materials with small endothermic gaps highly attractive for solar cell applications.

In conclusion, this Perspective discusses the unique aspects of singlet and triplet exciton transport in singlet fission materials. Specifically, as a result of interconversion between singlet and triplet population, triplet transport can be enhanced by singlet-mediated pathways, leading to long-range triplet transport in the nanosecond time scale. Cooperative single−triplet transport makes singlet fission materials with small endothermic gaps highly attractive for solar cell applications. We suggest that the design guidelines for singlet fission materials need to take coupled singlet−triplet transport into account in addition to optimize singlet fission rates.

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